



MMS050 Finite Element Simulation In Design

Hand in Assignment 1

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MMS050 Finite Element Simulation In Design

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1 Task 1: Design of a 2D truss bridge

1.1 Sub Task

I have considered the Hollow tube and cross section are outer diameter = 72mm, inner diameter = 50mm.

The total deformation and the normal force distribution is calculated. At 15 deg the compressive stress acting is 121.21 Mpa and FOS is 2.0

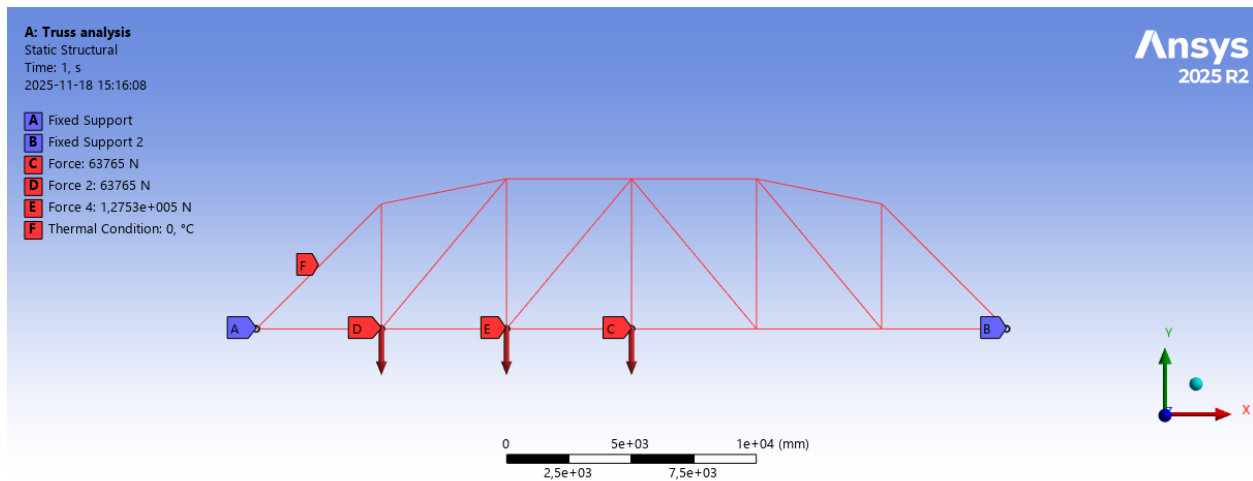


Figure 1: Force acting

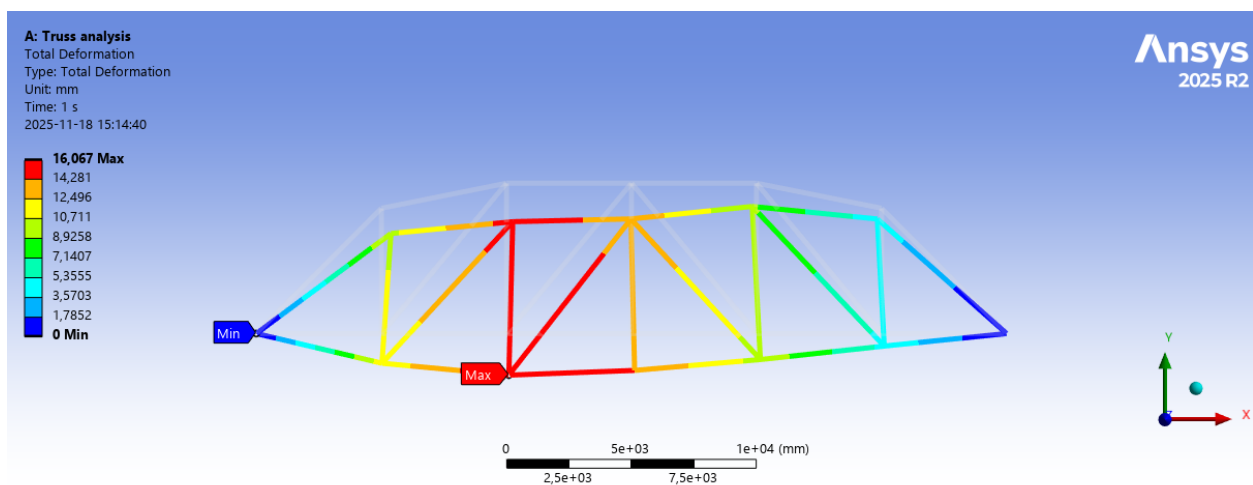


Figure 2: Total deformation at 15 deg

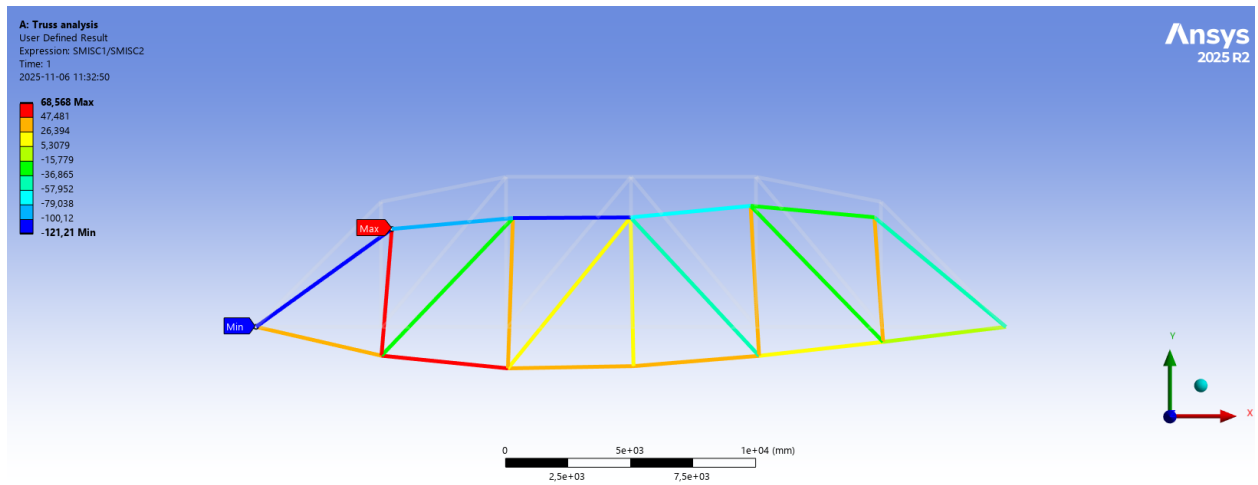


Figure 3: Normal stress at 15 deg

1.2 SubTask

At 50 deg compressive stress acting is 125.59 Mpa and Fos is 1.99

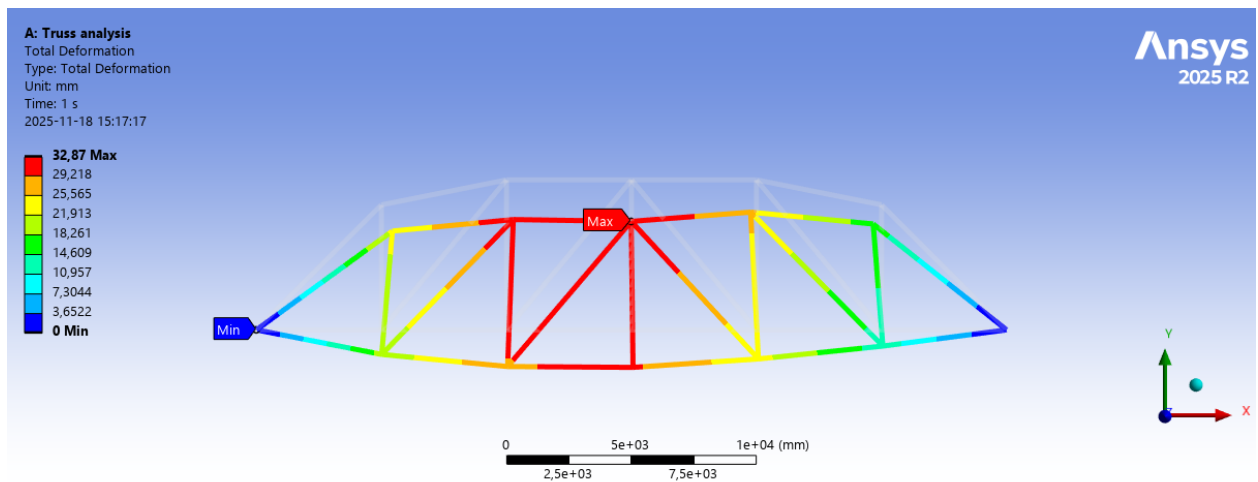


Figure 4: total deformation at 50 deg

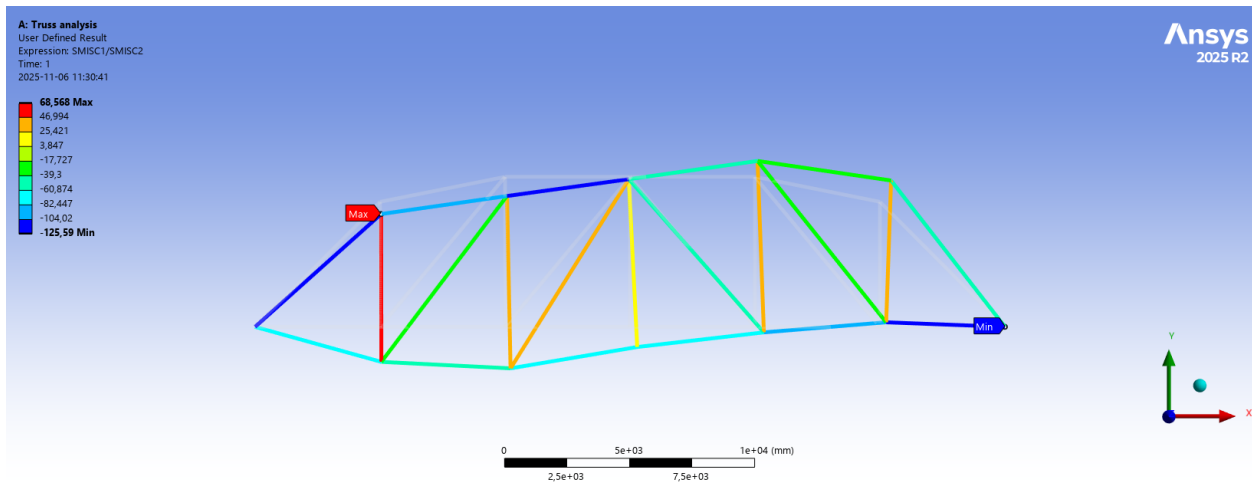


Figure 5: Normal stress at 50 deg

2 Task 2: Yandhai Nepean crossing

2.1 SubTask

The number of people can stand on the bridge at the same time is calculated as = Applied force(N) / weight per person(Kg) * gravity(m/s²).

$10400000 / 80 * 9.81 = 13,251$ people.

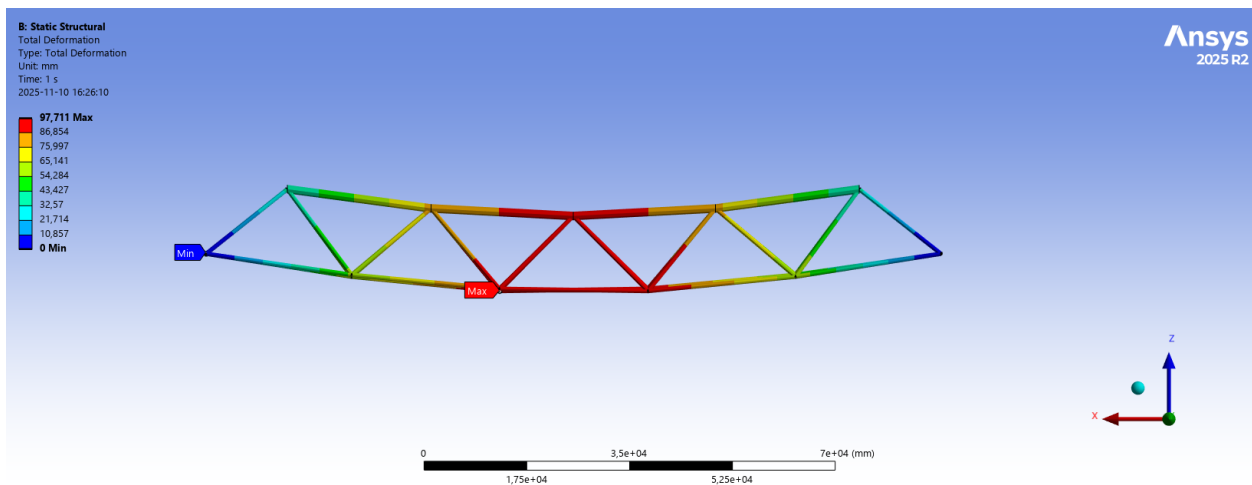


Figure 6: Total Deflection

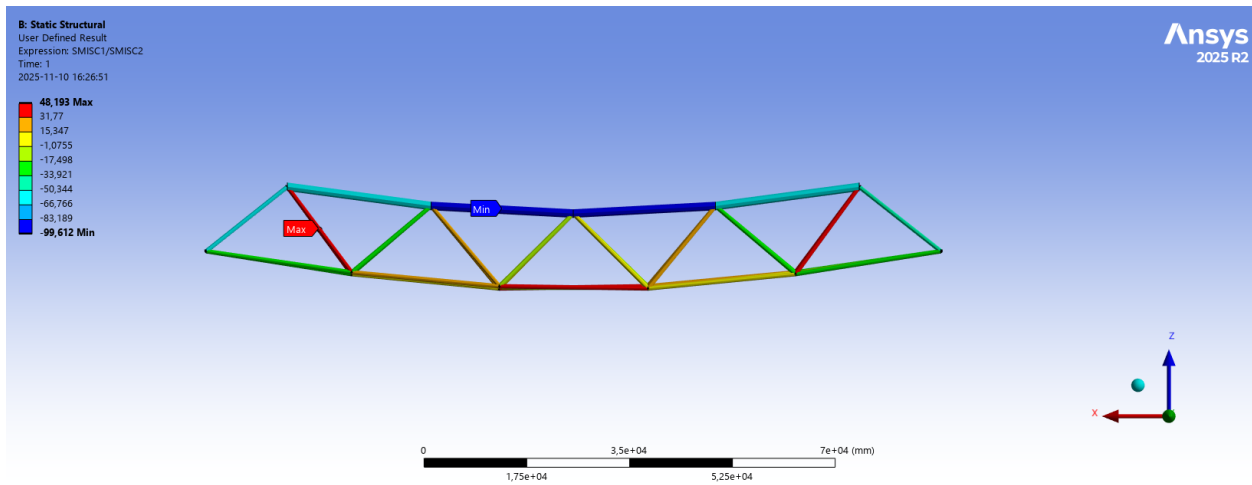


Figure 7: Normal stress

2.2 SubTask

I have considered the car of wheel base 2500mm because of this i have choosen this specific nodes as vehicle load is distributed on this 2 vertices.

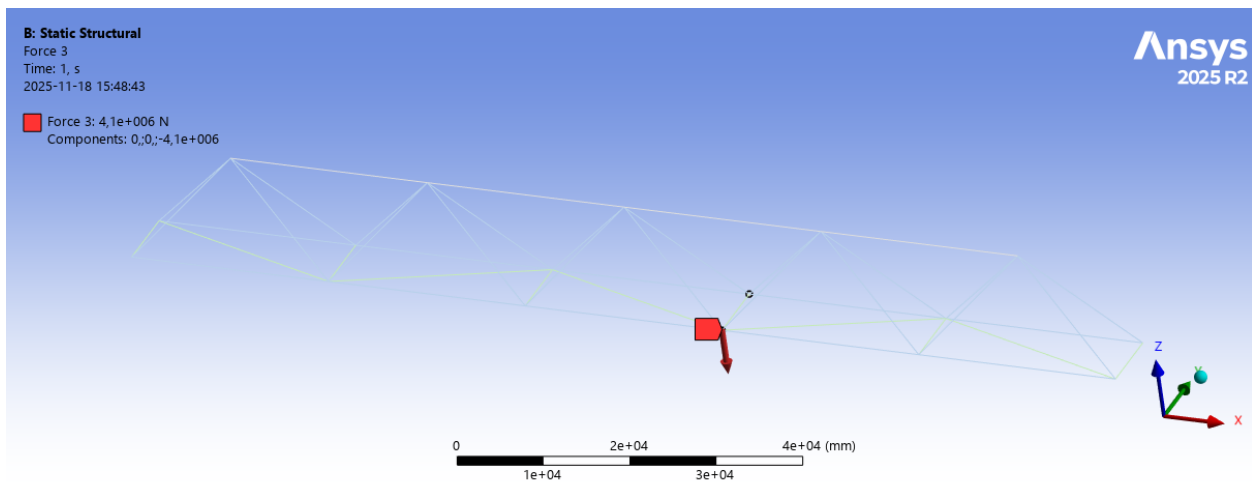


Figure 8: Vehicle load distributed at this 2 vertices

The vehicle weight acting on the bridge is = Applied force (N) / Gravity (m/s^2)

$$4100000 \text{ (N)} / 9.81 \text{ (m/s}^2\text{)} = 417940.87 \text{ Kg}$$

The maximum deflection and stress of the bridge is below 100mm and 100 Mpa.

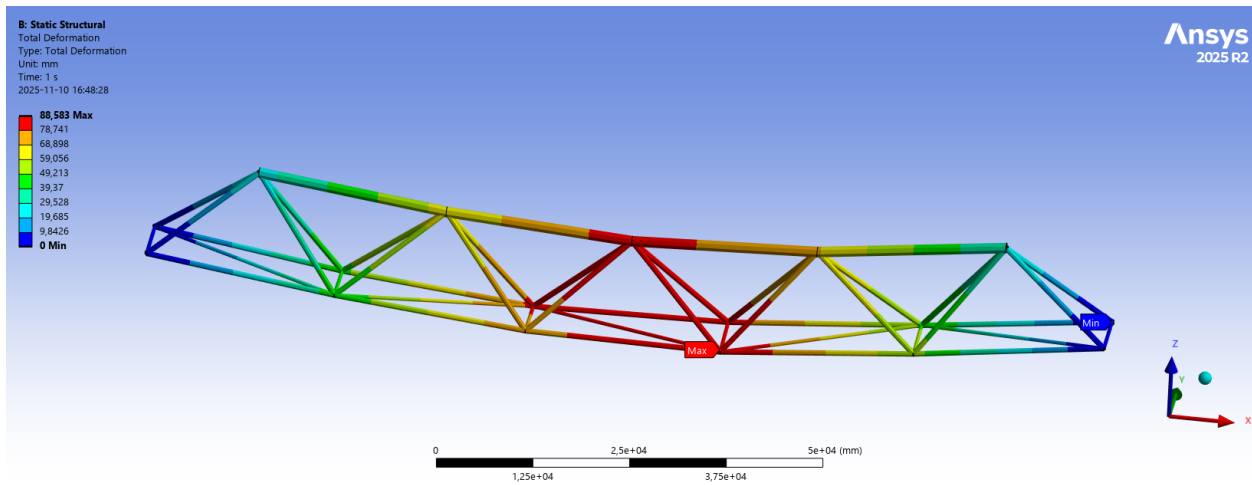


Figure 9: Total deformation due to vehicle weight

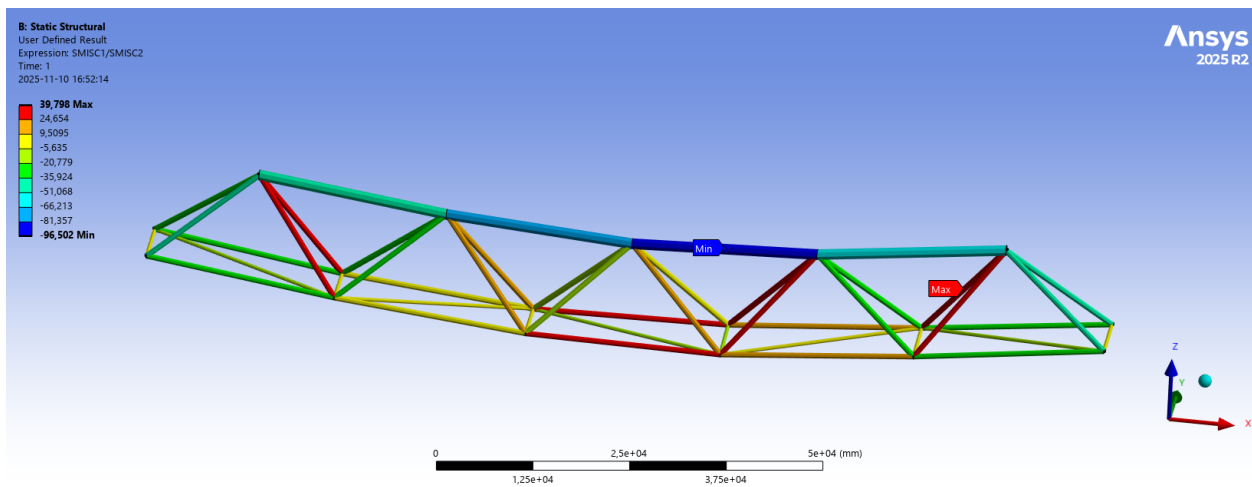


Figure 10: Normal stress due to vehicle weight

3 Task 3: Gear and rack

3.1 SubTask

The total deformation and von mises stress for gear and rack at thickness = 12mm, is shown below fig 11 and fig 12.

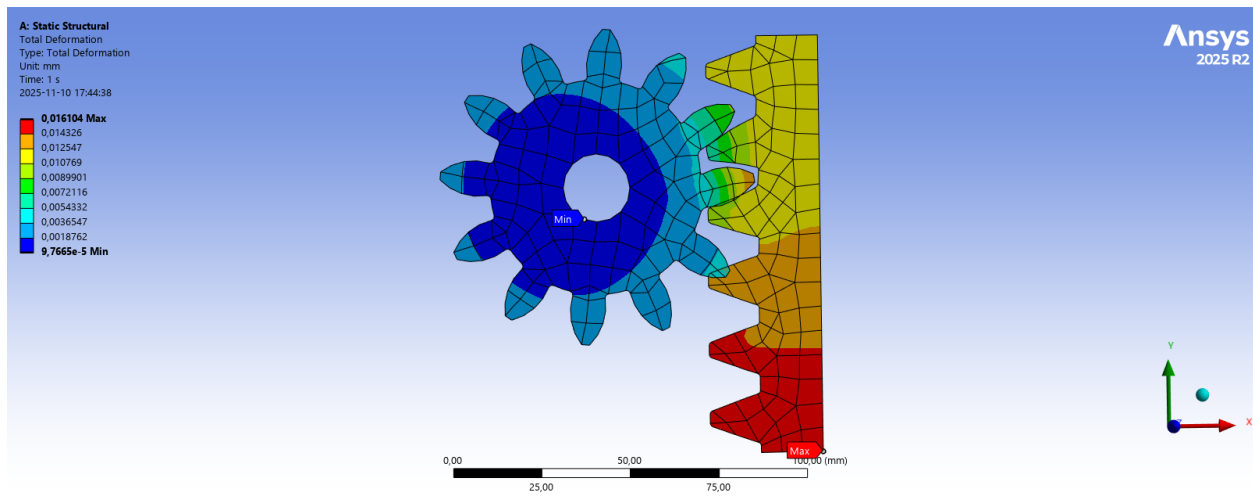


Figure 11: Total deformation of gear and rack at thickness 12mm

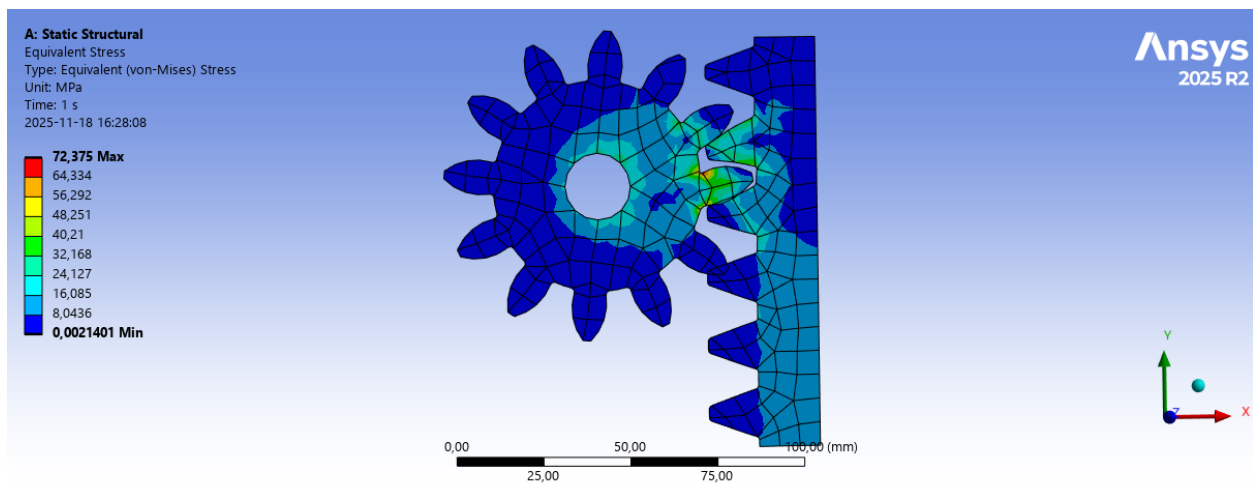


Figure 12: Von mises stress of gear and rack at thickness 12mm

3.2 SubTask

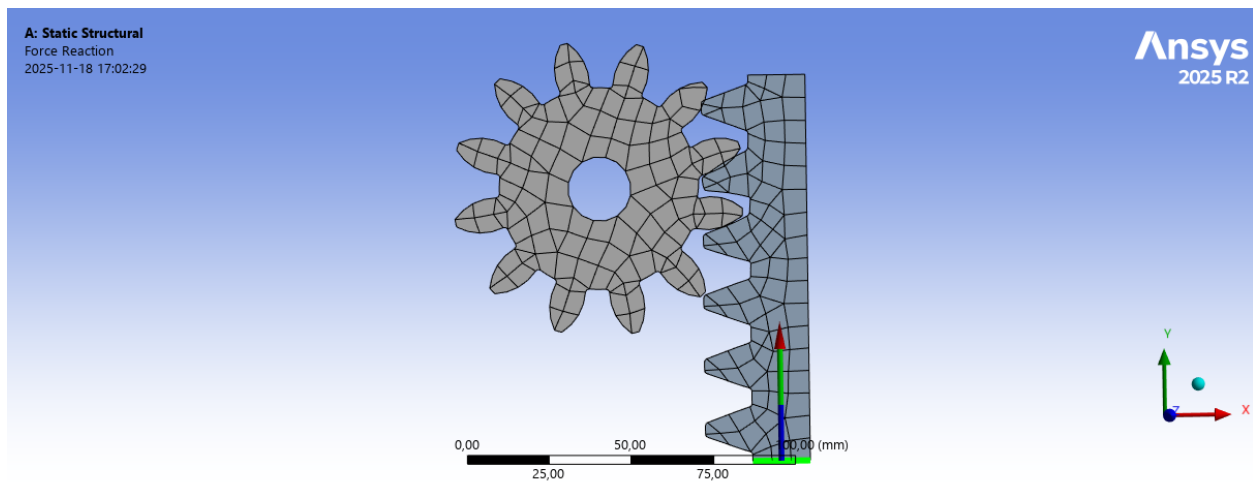


Figure 13: Force Reaction

The 2500N force reaction will be reduced to 2492N. Refer fig13 for the force reaction.

3.3 SubTask

In this model finer mesh of 1mm is applied and the thickness of gear and rack is chosen to be 13mm. For this thickness factor of safety is around 2.04.

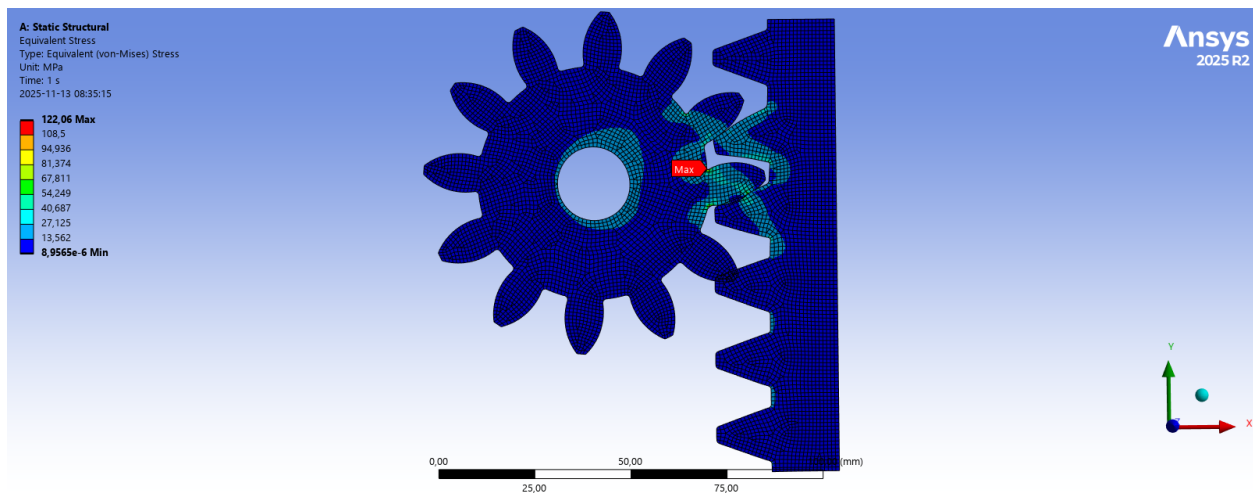


Figure 14: Von mises stress at thickness 13mm

4 Task 4: Analysis of an impeller pump

4.1 SubTask

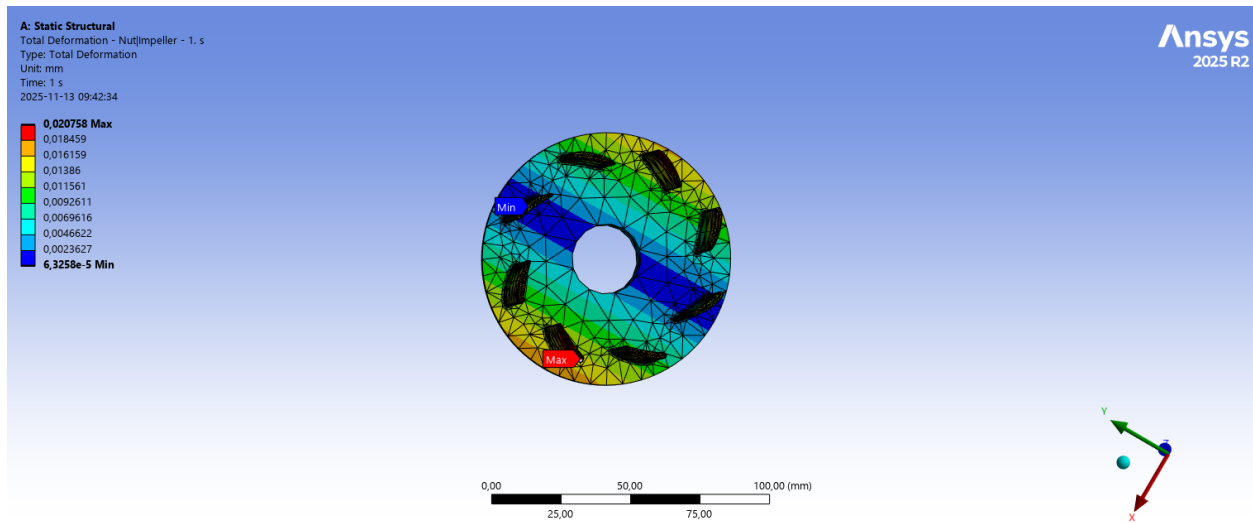


Figure 15: Total deformation of impeller

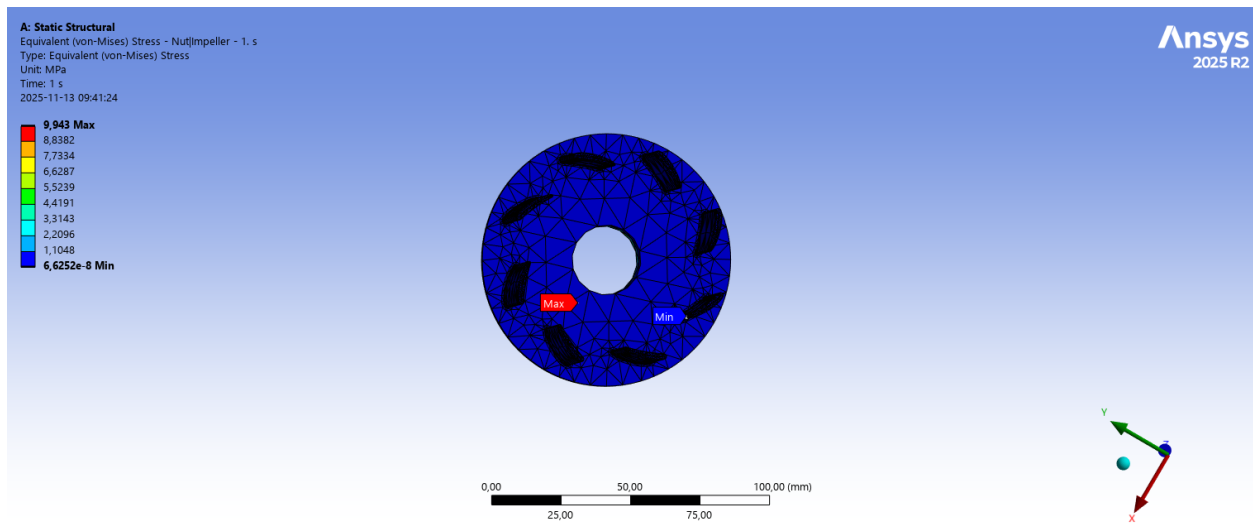


Figure 16: Von Mises stress of impeller

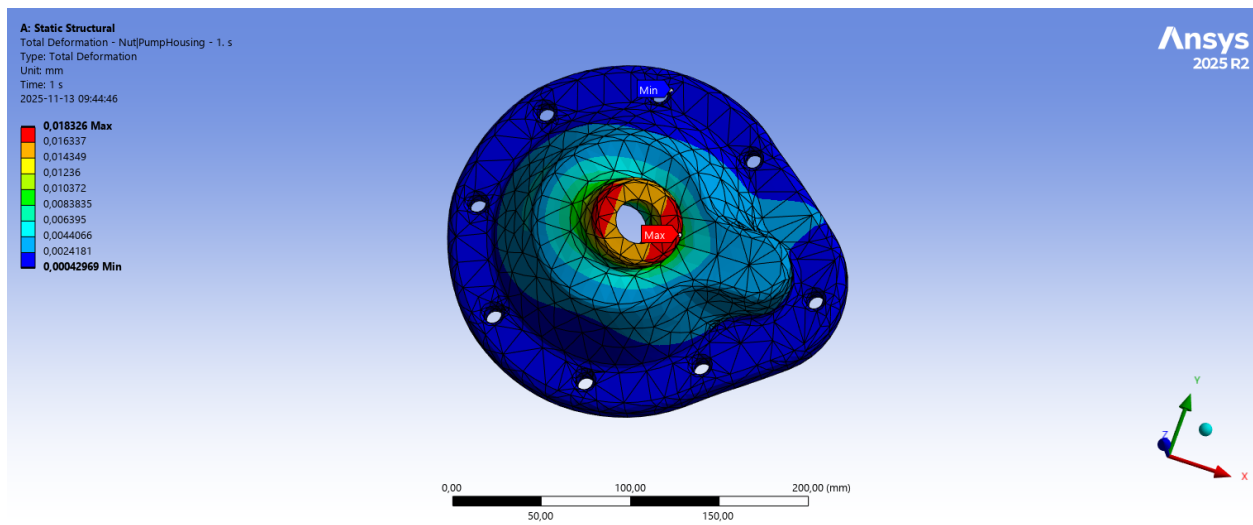


Figure 17: Total deformation of Pump housing

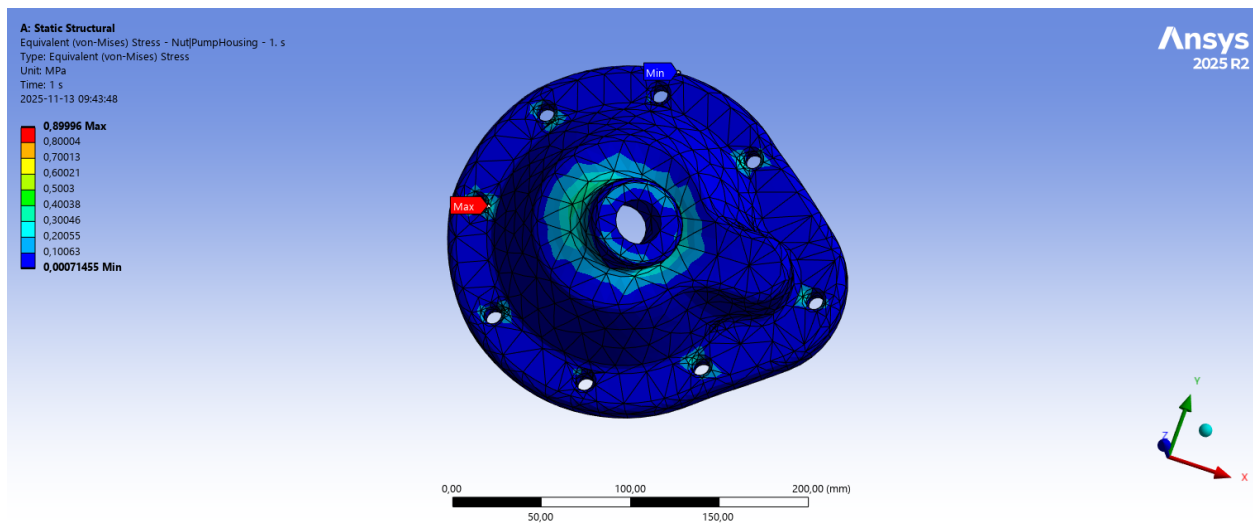


Figure 18: Von Mises stress of Pump housing

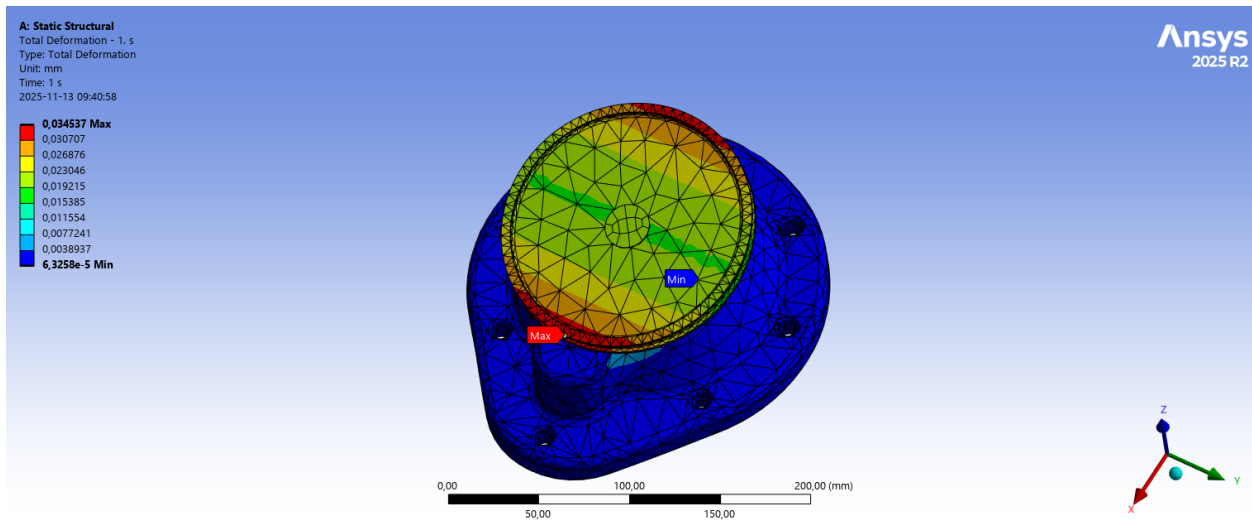


Figure 19: Total deformation of impeller pump assy

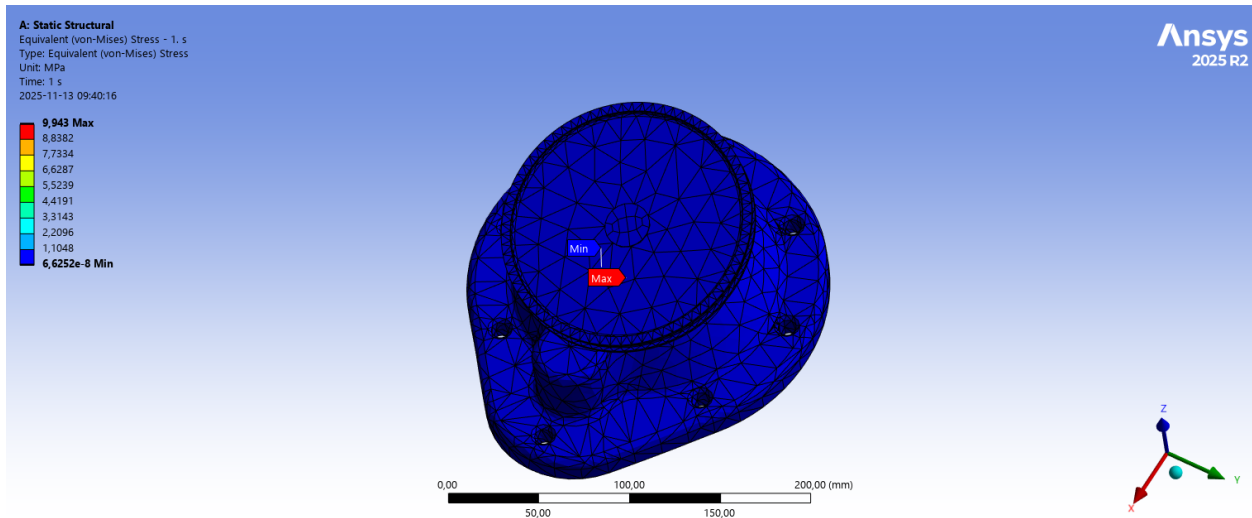


Figure 20: Von Mises stress of impeller pump assy

The total deformation and von Mises stress for the Impeller, Pump housing and the impeller pump assy indicating their maximum stress and location is shown in the figures above.

4.2 SubTask

Factor of Safety(FOS) for both impeller and pump housing is calculated.

Parts	Material	Stress(Mpa)	FOS	Deformation(mm)	FOS
Impeller	Structural steel	9.943	25.14	0.020758	3.62
Pump housing	Polyethylene	0.8999	10	0.018326	4.09

Table 1: FOS for impeller and pump housing

The deformation governs the design as it FOS is lower 3.62.

4.3 SubTask

In order to increase the FOS from task 4.2, I would look for material with higher yield strength so that it can withstand greater loads.

4.4 SubTask

The averaged contours make it easier to spot high-stress region and interpret results and useful for overall assessment of stress distribution. Unaveraged results shows true computed stresses, which have discontinuities at element boundaries.

Conclusion - For critical design decisions, failure analysis, always examine unaveraged stress values to detect true local maxima and realistic discontinuities.

5 Task 5: Parameter management

5.1 SubTask

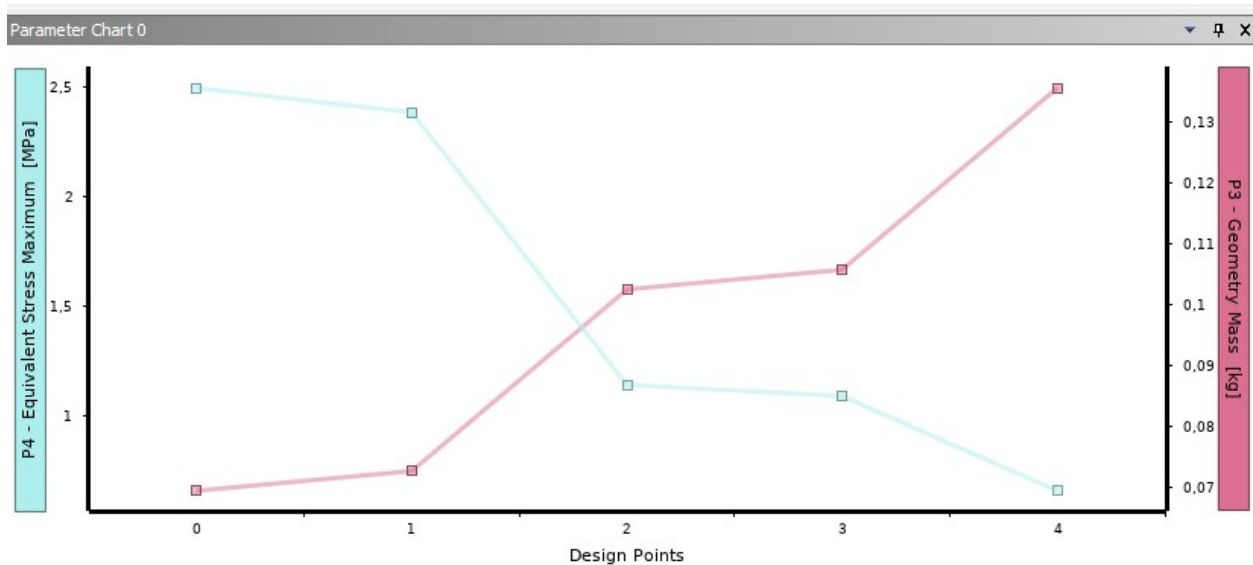


Figure 21: Design point vs output parameters)

5.2 SubTask

Table of Design Points						
	A	B	C	D	E	F
1	Name	P1 - SYS\Gusset Thickness	P2 - SYS\Bracket Thickness	P3 - Geometry Mass	P4 - Equivalent Stress Maximum	Retain
2	Units	mm	mm	kg	MPa	
3	DP 0 (Current)	1,5	2,5	0,08755	7,105	☒
4	DP 1	2,5	2,5	0,090788	6,868	☐
5	DP 2	1,5	3,5	0,12063	3,7003	☐
6	DP 3	2,5	3,5	0,12386	3,5728	☐
7	DP 4	1,5	4,5	0,1537	2,3114	☐
*						☐

Figure 22: New Thickness with output parameter

I am recommending a design with a bracket thickness of 2.5 mm and a gusset thickness of 1.5 mm, ensuring that the maximum stress remains below 12 MPa and the maximum mass does not exceed 0.10 kg.

Geometry mass is 0.08755 kg.

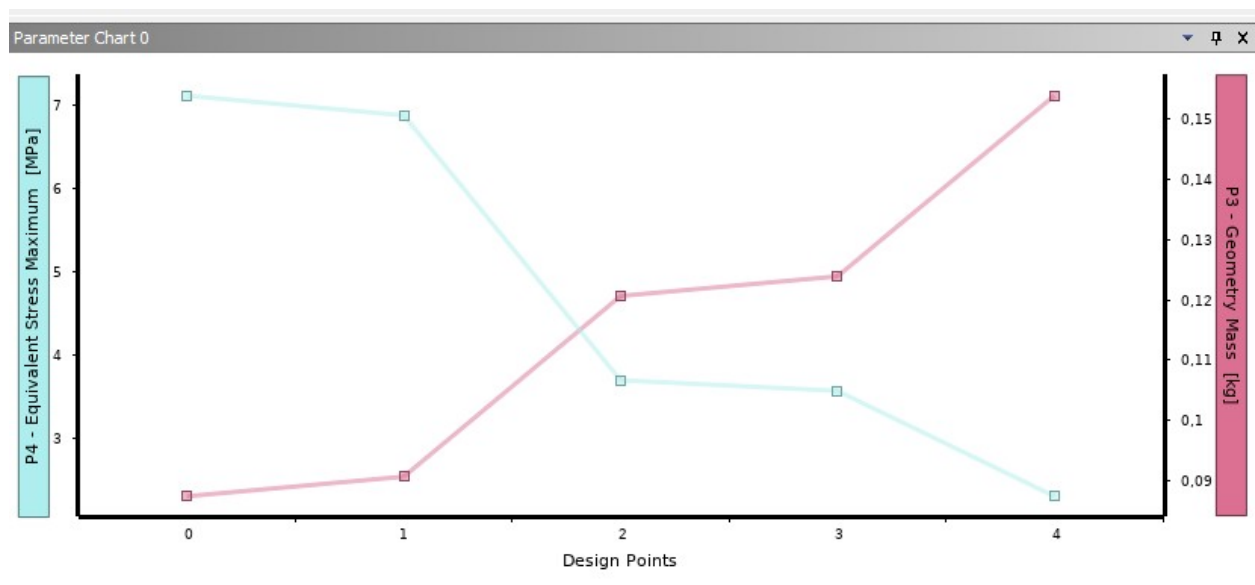


Figure 23: Design points vs output parameter

Stress is 11.231Mpa

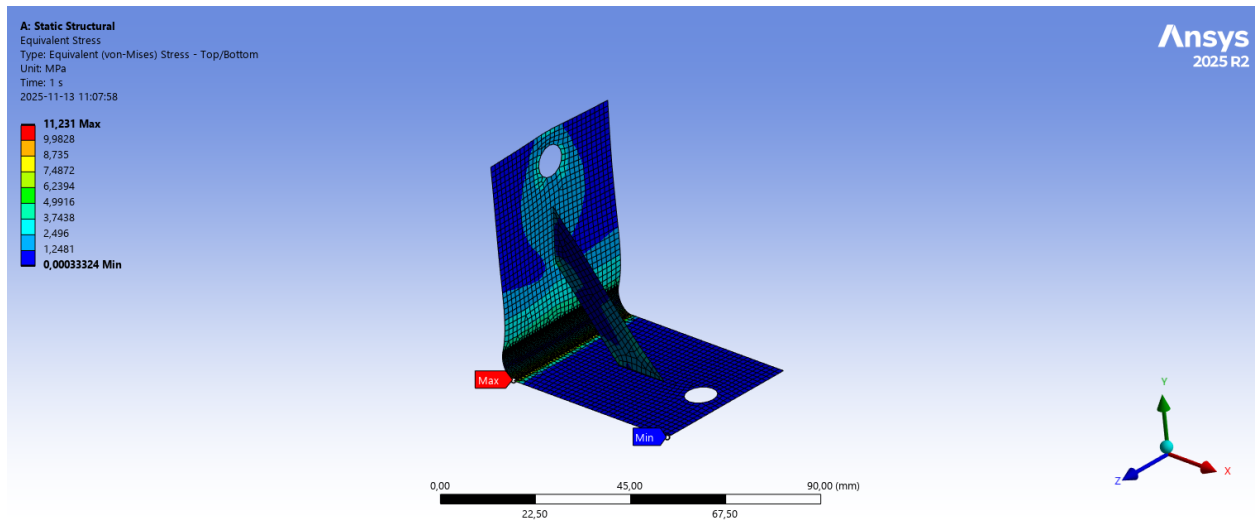


Figure 24: Surface plot of the von Mises stress